

Advanced Mathematical Techniques

Bookwork 4

1. **Problem 5.9.12:** Show that:

$$\int_0^\infty \frac{(\ln z)^2}{1+z^2} dz = 4 \left(1 - \frac{1}{3^3} + \frac{1}{5^3} - \frac{1}{7^3} + \cdots \right)$$

$$\begin{aligned} \int_0^\infty \frac{(\ln z)^2}{1+z^2} dz &= \int_0^1 \frac{(\ln z)^2}{1+z^2} dz + \int_1^\infty \frac{(\ln z)^2}{1+z^2} dz = 2 \int_0^1 \frac{(\ln x)^2}{1+x^2} dx = 2 \sum_{n=0}^\infty (-1)^n \int_0^1 x^{2n} \ln^2 x dx \\ &= 2 \sum_{n=0}^\infty (-1)^n \int_0^1 x^{2n} \ln^2 x dx = 4 \sum_{n=0}^\infty \frac{(-1)^n}{(2n+1)^3} = 4 \left(1 - \frac{1}{3^3} + \frac{1}{5^3} - \frac{1}{7^3} + \cdots \right) \checkmark \end{aligned}$$

2. **Problem 8.4.6:** Show, by means of the beta function, that:

3.

$$\int_t^z \frac{dx}{(z-x)^{1-\alpha}(x-t)^\alpha} = \frac{\pi}{\sin \pi \alpha}, 0 < \alpha < 1$$

$$\begin{aligned} x-t &= u(z-t), z-x = (z-t)(1-u), \int_0^1 \frac{(z-t)du}{((z-t)(1-u))^{1-\alpha}(u(z-t))^\alpha} = \int_0^1 u^{-\alpha}(1-u)^{\alpha-1} \\ &= \Gamma(1-\alpha)\Gamma(\alpha) = \boxed{\frac{\pi}{\sin \pi \alpha}} \checkmark \end{aligned}$$

4. **Problem 8.4.10:** Evaluate $\int_0^1 (1-x^4)^{-\frac{1}{2}} dx$ as a beta function:

$$t = x^4, \int_0^1 (1-t)^{-\frac{1}{2}} \cdot \frac{1}{4} \cdot t^{-\frac{3}{4}} dt = \frac{1}{4} B\left(\frac{1}{4}, \frac{1}{2}\right) = \frac{1}{4} \frac{\Gamma(\frac{1}{4})\Gamma(\frac{1}{2})}{\Gamma(\frac{3}{4})} = \boxed{\frac{[(\frac{1}{4})!]^2 \cdot 4}{\sqrt{2\pi}}}$$